Symbol	Unit	Meaning
Indices		
$i,\ o$	Token	Input (sold) and output (bought) endpoint tokens, respectively; (i,o) is an ordered input–output pair and the direct route is $i \to o$.
x	Token	Generic token index used where a quantity applies to any endpoint or candidate token.
k	Token	Candidate vehicle token used as the route intermediate; $k \notin \{i,o\}$.
h	Token	Candidate vehicle token used as a challenger to incumbent k; $h \in \mathcal{K} \setminus \{i,o,k\}$.
$\ell,\ p,\ p'$	Token / pool	ℓ indexes tokens in cross-token sums; p indexes a focal liquidity pool; p' indexes another pool in leave-one-out sums.
$a,\ a'$	LP address	a indexes a focal liquidity-provider address or resolved position controller; a' indexes another provider in within-pool sums.
b	Fraction	Symmetric price-band half-width used to measure concentrated liquidity around the current reference price; $b = 0.02$ is primary.
$t,\ u,\ w$	UTC day / UTC week	t indexes a focal calendar day; u indexes another calendar day in a time-window sum; w indexes calendar weeks in settlement variables.
τ	Days	Positive calendar-day horizon selected ex ante for each dynamic specification.
$d,\ \mu$	Calendar days / calendar months	d is calendar-day event time relative to a candidate-specific shock; μ is calendar-month event time relative to an architecture activation date. Event time zero contains the event or activation date.
g	Settlement comparison cell	Settlement comparison cell defined by ordered pair (i,o), vehicle k, UTC week w, and a prespecified route-size bin.
q	USD	Input quote notional. Un-suffixed route-cost columns use $q = \$10{,}000$.
r	Route unit	Reconstructed input-to-output route unit. A coherent $i \to k \to o$ component contributes one r regardless of its number of legs; a split or join contributes one r per reconstructed input–output pair.
Route and liquidity aggregates		
$\mathrm{Vol}_t,\ \mathrm{DVol}_t$	USD	Vol_t is total realized USD volume across all direct and indirect route units on day t. DVol_t restricts it to direct routes, so $0 \le \mathrm{DVol}_t \le \mathrm{Vol}_t$.
$\mathrm{IVol}_t,\ \mathrm{IVol}_{k,t}$	USD	IVol_t restricts Vol_t to indirect routes, so $0 \le \mathrm{IVol}_t \le \mathrm{Vol}_t$. $\mathrm{IVol}_{k,t}$ further restricts that volume to routes using vehicle k, so $0 \le \mathrm{IVol}_{k,t} \le \mathrm{IVol}_t$.
$\mathrm{Vol}_{i,o,t},\ \mathrm{IVol}_{i,o,t},\ \mathrm{IVol}_{i,o,k,t}$	USD	$\mathrm{Vol}_{i,o,t}$ is realized route-unit volume for ordered pair (i,o) on day t. $\mathrm{IVol}_{i,o,t}$ restricts it to indirect routes, and $\mathrm{IVol}_{i,o,k,t}$ further restricts it to routes through k, so $0 \le \mathrm{IVol}_{i,o,k,t} \le \mathrm{IVol}_{i,o,t} \le \mathrm{Vol}_{i,o,t}$.
$N_t,\ N^{\mathrm{in}}_{k,t},\ N^{\mathrm{out}}_{k,t}$	Route-unit count	N_t counts all reconstructed route units r on day t, not their individual legs. $N^{\mathrm{in}}_{k,t}$ and $N^{\mathrm{out}}_{k,t}$ restrict that count to routes with k as the input or output endpoint, respectively; each lies between 0 and N_t.
$N^I_t,\ N^I_{k,t}$	Route-unit count	N^I_t restricts N_t to route units with at least one intermediate token. $N^I_{k,t}$ further restricts that count to units with intermediate k, so $0 \le N^I_{k,t} \le N^I_t \le N_t$; superscript I denotes indirect routes.
$\mathcal{A}_t,\ \mathcal{A}^k_t,\ \mathcal{M}^k_t$	Sets of token pairs	$\mathcal{A}_t$ is the set of endpoint pairs active in indirect routes on day t. $\mathcal{A}^k_t$ restricts it to pairs using k, and $\mathcal{M}^k_t$ further restricts it to pairs for which k has the largest candidate-vehicle volume; $\mathcal{M}^k_t \subseteq \mathcal{A}^k_t \subseteq \mathcal{A}_t$.
$\mathrm{LegVol}^{\mathrm{in}}_{k,t},\ \mathrm{LegVol}^{\mathrm{out}}_{k,t}$	USD	$\mathrm{LegVol}^{\mathrm{in}}_{k,t}$ and $\mathrm{LegVol}^{\mathrm{out}}_{k,t}$ are route-leg USD volumes for which k is the leg's input or output token, respectively, on day t.
$\mathrm{Vol}^{\mathrm{in}}_{k,t},\ \mathrm{Vol}^{\mathrm{out}}_{k,t}$	USD	Input- and output-endpoint restrictions of Vol_t for token k: $0 \le \mathrm{Vol}^{\mathrm{in}}_{k,t} \le \mathrm{Vol}_t$ and $0 \le \mathrm{Vol}^{\mathrm{out}}_{k,t} \le \mathrm{Vol}_t$.
$\mathcal{K}$	Set of tokens	Prespecified candidate set $\{\mathrm{WETH}, \mathrm{USDC}, \mathrm{USDT}, \mathrm{DAI}, \mathrm{WBTC}\}$.
$\mathcal{L}_t,\ \mathcal{L}_{k,t},\ m_p$	Set of pools / token count	$\mathcal{L}_t$ contains pools from protocols whose deposited-capital contract is admitted, with exact token contracts and $0 < \mathrm{Capital}_{p,t} \le 10$ billion USD. $\mathcal{L}_{k,t} \subseteq \mathcal{L}_t$ restricts that set to pools with candidate k on one side. m_p is the number of pool tokens in $\mathcal{K}$, so $m_p \in \{1,2\}$.
$\mathrm{Capital}_{p,t},\ C_{k,t}$	USD	$\mathrm{Capital}_{p,t}$ is pool p's validated day-t accounting capital under its protocol contract. $C_{k,t} = \sum_{p \in \mathcal{L}_{k,t}} \mathrm{Capital}_{p,t}/m_p$; a one-candidate pool contributes all capital to that candidate and a two-candidate pool contributes one half to each.
$\mathrm{PoolVol}_{p,t},\ \mathrm{SpokeIVol}_{p,k,t}$	USD	$\mathrm{PoolVol}_{p,t}$ is total realized swap volume in pool p on day t. $\mathrm{SpokeIVol}_{p,k,t}$ restricts it to pool legs belonging to reconstructed indirect routes that use k as the intermediate, so $0 \le \mathrm{SpokeIVol}_{p,k,t} \le \mathrm{PoolVol}_{p,t}$.
LP portfolio objects		
$L_{a,p,t}$	USD	End-of-day USD value of active liquidity supplied by provider a to pool p, marked at the day-t reference prices.
$w_{a,p,t}$	Fraction (0–1)	Provider a's share of active capital in pool p on day t: $w_{a,p,t} = L_{a,p,t}/\sum_{a'} L_{a',p,t}$.
$F^{\mathrm{LP}}_{a,p,t}$	USD per day	Day-t liquidity deposits by a into p minus withdrawals, valued at transaction-time USD prices and excluding fee collections.
$\mathrm{Fee}_{a,p,t},\ G^{\mathrm{LP}}_{a,p,t}$	USD	Swap fees accrued to provider a in pool p during day t, and USD gas spent by a on that position's mint, burn, collect, and rebalance transactions.
$V^{\mathrm{LP}}_{a,p,t},\ V^{\mathrm{RB}}_{a,p,t}$	USD	End-of-day values of the day-t opening LP inventory and its self-financing rebalanced benchmark, respectively, both excluding day-t fees, flows, and gas.
$R^{\mathrm{LP}}_{a,p,t},\ R^{\mathrm{other}}_{a,-p,t}$	Daily return fraction	$R^{\mathrm{LP}}_{a,p,t} = (V^{\mathrm{LP}}_{a,p,t} - L_{a,p,t-1})/L_{a,p,t-1}$ is the position's fee-, flow-, and gas-excluded return. $R^{\mathrm{other}}_{a,-p,t}$ is the lag-capital-weighted mean of that return across a's pools other than p.
$\omega_{a,x,-p,t}$	Fraction (0–1)	Share of provider a's marked LP inventory outside focal pool p exposed to token x, excluding both tokens in p and normalized to sum to one across the remaining tokens.
$\mathrm{BandDepth}_{p,t,b}$	USD	Average across trade directions of the input USD notional executable in pool p before its marginal price exits the symmetric b band around the day-t reference price.
Route and liquidity aggregates		
R^{WETH}_t	Daily log-return fraction	Day-t log return of the WETH price used to construct downside stress.
Candidate-shock objects		
$P_{x,t},\ P_{k,t}$	USD per token	$P_{x,t}$ is the day-t USD price of any token x. $P_{k,t}$ restricts that price to candidate $k \in \mathcal{K}$.
$R_{x,t},\ R_{k,t}$	Daily log-return fraction	$R_{x,t} = \ln(P_{x,t}/P_{x,t-1})$ is the day-t return of any token x. $R_{k,t}$ restricts it to candidate $k \in \mathcal{K}$.
$\sigma^{(30)}_{k,t-1}$	Daily log-return standard deviation	Sample standard deviation of $R_{k,u}$ over the 30 calendar days ending at $t-1$, requiring at least 20 valid daily returns.
Architecture objects		
$R_{i,o,t},\ \sigma^{\mathrm{pre}}_{i,o}$	Daily log-return fraction / daily log-return standard deviation	$R_{i,o,t} = \ln[(P_{i,t}/P_{o,t})/(P_{i,t-1}/P_{o,t-1})]$ is the ordered endpoint-pair return. $\sigma^{\mathrm{pre}}_{i,o}$ is its sample standard deviation over $\mathcal{T}^{\mathrm{V3}}_{\mathrm{pre}}$.
Persistence and displacement objects		
$k^\star_{i,o,t},\ h^\star_{i,o,q,t}$	Token	$k^\star_{i,o,t}$ is the incumbent vehicle with the largest mean $\mathrm{VehicleShare}_{i,o,k,u}$ over the 30 calendar days ending at $t-1$. $h^\star_{i,o,q,t}$ is the executable nonincumbent candidate with the smallest all-in indirect cost $C^I_{i,o,h,q,t}$ on day t.
Architecture objects		
$t^{\mathrm{V3}}_0,\ \mathcal{T}^{\mathrm{V3}}_{\mathrm{pre}},\ \mathcal{P}^{\mathrm{V3}}_q$	UTC day / set of UTC days / set of token pairs	t^{V3}_0 is the Uniswap V3 activation date, 5 May 2021. $\mathcal{T}^{\mathrm{V3}}_{\mathrm{pre}}$ is the fixed 180-calendar-day window ending at $t^{\mathrm{V3}}_0 - 1$. $\mathcal{P}^{\mathrm{V3}}_q$ is the fixed architecture-analysis universe of ordered pairs with positive realized route volume on at least 30 days in that pre-period; every member is quoted at q throughout the event window, independent of post-V3 activity.
$t^{\mathrm{V4}}_0,\ \mathcal{T}^{\mathrm{V4}}_{\mathrm{pre}},\ \mathcal{T}^{\mathrm{V4}}_{i,o,\mathrm{pre}}$	UTC day / sets of UTC days	t^{V4}_0 is the Ethereum V4 activation date established from the canonical deployment metadata before extraction. $\mathcal{T}^{\mathrm{V4}}_{\mathrm{pre}}$ is the fixed 180-calendar-day window ending at $t^{\mathrm{V4}}_0 - 1$. $\mathcal{T}^{\mathrm{V4}}_{i,o,\mathrm{pre}} \subseteq \mathcal{T}^{\mathrm{V4}}_{\mathrm{pre}}$ restricts that window to days with $\mathrm{Vol}_{i,o,t} > 0$.
Route-cost quote objects		
$\mathcal{P}_{k,t,q}$	Set of token pairs	Quote-universe pairs for candidate k: distinct ordered input–output pairs among day t's 200 largest clean reconstructed pairs by realized USD volume, where clean means route class `single` or `coherent`, $k \notin \{i,o\}$, and each of i, o, and k has a valid day-price estimate. A valid estimate requires at least three finite token-side USD-per-token observations in $(0, \$1{,}000{,}000)$, at least 75 percent of observations within a fivefold band around their ordinary median, and equals the realized-USD-volume-weighted median inside that consensus band. Each pair is submitted to the direct and via-k quote engines at input q; membership does not require either quote to execute.
$\mathcal{D}_{k,t,q},\ \mathcal{I}_{k,t,q}$	Sets of token pairs	$\mathcal{D}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ and $\mathcal{I}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ restrict the broad pair set to pairs with an executable direct route or an executable indirect route via k.
$\mathcal{C}_{k,t,q}$	Set of token pairs	Common-support pairs for candidate k, day t, and notional q: $\mathcal{C}_{k,t,q} = \mathcal{D}_{k,t,q} \cap \mathcal{I}_{k,t,q}$, so both the direct and indirect routes are executable.
$\mathcal{T}_{k,t,q},\ \mathcal{W}_{k,t,q}$	Sets of token pairs	$\mathcal{T}_{k,t,q} \subseteq \mathcal{D}_{k,t,q}$ is the thin-direct subset. $\mathcal{W}_{k,t,q} \subseteq \mathcal{C}_{k,t,q}$ contains common-support pairs for which the indirect route beats the direct route, equivalently $\Delta C^D_{i,o,k,q,t} < 0$.
$D_{i,o,q,t},\ I_{i,o,k,q,t},\ T_{i,o,q,t}$	Indicator (0/1)	Indicators for direct-route availability (D), indirect-route-through-k availability (I), and an executable direct route returning less than $0.9q$ (T).
$O^D_{i,o,q,t},\ O^I_{i,o,k,q,t}$	USD	Quoted output values; superscripts D and I denote direct and indirect routes.
$O^{D,\mathrm{fee}}_{i,o,q,t},\ O^{I,\mathrm{fee}}_{i,o,k,q,t}$	USD	Counterfactual direct and indirect output values when each pool remains at its pre-quote marginal price but charges its historical swap fee; these retain fee loss and suppress price impact.
$G^D_{i,o,q,t},\ G^I_{i,o,k,q,t}$	USD	Historical gas expenditure for the quoted direct and indirect routes, respectively, using route-specific gas usage and the day-t gas-token USD price.
$C^D_{i,o,q,t},\ C^I_{i,o,k,q,t}$	Fraction	All-in execution-cost fractions for the direct and indirect alternatives, including quoted pool fees and price impact plus historical gas expenditure.
$\Delta C^D_{i,o,k,q,t}$	Fraction	Pair-level direct cost advantage $(O^D_{i,o,q,t} - O^I_{i,o,k,q,t})/O^D_{i,o,q,t}$ on $\mathcal{C}_{k,t,q}$; positive values favor the direct route.
$\Delta C^{D,\mathrm{all}}_{i,o,k,q,t}$	Fraction	All-in direct cost advantage $C^I_{i,o,k,q,t} - C^D_{i,o,q,t}$ on $\mathcal{C}_{k,t,q}$; positive values favor the direct route.
Settlement objects and operators		
$\mathcal{R}_{k,w}$	Set of route units	Receipt-audited matched route units using vehicle k in UTC week w.
$\mathcal{R}^{\mathrm{transfer}}_{k,w}$	Set of route units	Members whose receipt logs a transfer of vehicle k; $\mathcal{R}^{\mathrm{transfer}}_{k,w} \subseteq \mathcal{R}_{k,w}$.
$\mathcal{R}^3_g,\ \mathcal{R}^4_g$	Sets of route units	Receipt-audited route units in settlement comparison cell g executed on Uniswap V3 and V4, respectively. A matched cell has both sets nonempty.
$M_{r,k},\ \mathrm{GrossLegVol}_{r,k}$	USD	$M_{r,k}$ is the sum of absolute, route-attributed ERC-20 transfers of intermediate k between distinct nonzero addresses in route unit r, valued in USD. $\mathrm{GrossLegVol}_{r,k}$ is the corresponding gross USD notional of k entering or leaving the route's pools across both vehicle legs.
$\lvert\mathcal{S}\rvert,\ \mathbf{1}_{\{\cdot\}}$	Count / indicator (0/1)	Cardinality of any finite set $\mathcal{S}$ and an indicator equal to one when its subscripted condition is true.
Dynamic operators		
Δ_τ		Change in a daily variable over the τ days ending at t: $\Delta_\tau X_t = X_t - X_{t-\tau}$.

Variable	Formula	Unit	Data column	Definition
Vehicle-use measures				
$\mathrm{VehicleShare}_{k,t}$	$\frac{\mathrm{IVol}_{k,t}}{\mathrm{IVol}_t}$	Fraction (0–1)	`bridge_share`	Fraction of day-t indirect-route USD volume routed through k.
$\mathrm{AllRouteVehicleShare}_{k,t}$	$\frac{\mathrm{IVol}_{k,t}}{\mathrm{Vol}_t}$	Fraction (0–1)	`all_route_bridge_share`	Fraction of day-t all-route USD volume routed indirectly through k.
$\mathrm{IShare}_{k,t}$	$\frac{\mathrm{IVol}_{k,t}}{\sum_{j:\mathrm{Type}(j)\neq\mathrm{other}} \mathrm{IVol}_{j,t}}$	Fraction (0–1)	`intermediate_share`	Token k's share of clean, non-cyclic route value carried as an intermediary; a repeated appearance of k inside one reconstructed component is counted once. Primary shares use the prespecified currency types and exclude the residual unclassified-contract bucket.
$\mathrm{EShare}_{k,t}$	$\frac{\mathrm{EVol}_{k,t}}{\sum_{j:\mathrm{Type}(j)\neq\mathrm{other}} \mathrm{EVol}_{j,t}}$	Fraction (0–1)	`endpoint_share`	Token k's share of source-or-sink value across all clean, non-cyclic routes, including direct routes so the benchmark does not condition on intermediation. Primary shares use the same prespecified currency types.
$\mathrm{ExcessUse}_{k,t}$	$\frac{\mathrm{IShare}_{k,t}}{\mathrm{EShare}_{k,t}}$	Ratio	`vehicle_excess_use_ratio`	Intermediate-use share divided by endpoint-demand share on the same clean, non-cyclic route universe; undefined when endpoint demand is zero.
$\mathrm{IShare}^N_{k,t}$	$\frac{N^I_{k,t}}{\sum_{j:\mathrm{Type}(j)\neq\mathrm{other}} N^I_{j,t}}$	Fraction (0–1)	`intermediate_count_share`	Token k's share of clean, non-cyclic route components in which it is an intermediary; one component-token appearance is counted once.
$\mathrm{EShare}^N_{k,t}$	$\frac{N^E_{k,t}}{\sum_{j:\mathrm{Type}(j)\neq\mathrm{other}} N^E_{j,t}}$	Fraction (0–1)	`endpoint_count_share`	Token k's share of source-or-sink appearances across all clean, non-cyclic route components, including direct routes.
$\mathrm{ExcessUse}^N_{k,t}$	$\frac{\mathrm{IShare}^N_{k,t}}{\mathrm{EShare}^N_{k,t}}$	Ratio	`vehicle_excess_use_count_ratio`	Count-share analogue of vehicle excess use on the same clean, non-cyclic route universe; undefined when endpoint count is zero.
$\mathrm{VehicleCountShare}_{k,t}$	$\frac{N^I_{k,t}}{N^I_t}$	Fraction (0–1)	`bridge_count_share`	Fraction of day-t indirect route units that use k.
$\mathrm{PairCoverage}_{k,t}$	$\frac{\lvert\mathcal{A}^k_t\rvert}{\lvert\mathcal{A}_t\rvert}$	Fraction (0–1)	`pair_coverage`	Fraction of active indirect-route endpoint pairs that use k.
$\mathrm{MainVehiclePairShare}_{k,t}$	$\frac{\lvert\mathcal{M}^k_t\rvert}{\lvert\mathcal{A}_t\rvert}$	Fraction (0–1)	`pair_main_vehicle_share`	Fraction of active indirect-route endpoint pairs for which k has the largest candidate-vehicle volume.
$\mathrm{IVol}_{k,t}$		USD	`bridge_volume_usd`	Sum of realized USD volume over indirect routes using vehicle k on day t.
Pair-level route-choice measures				
$\mathrm{VehicleShare}_{i,o,k,t}$	$\frac{\mathrm{IVol}_{i,o,k,t}}{\mathrm{IVol}_{i,o,t}}$	Fraction (0–1)	`actual_vehicle_share`	Candidate k's share of realized indirect-route USD volume for ordered pair (i,o) on day t, defined when $\mathrm{IVol}_{i,o,t} > 0$.
$\mathrm{IndirectRouteShare}_{i,o,t}$	$\frac{\mathrm{IVol}_{i,o,t}}{\mathrm{Vol}_{i,o,t}}$	Fraction (0–1)	`pair_indirect_route_share`	Fraction of realized route-unit USD volume for ordered pair (i,o) executed through an indirect route on day t, defined when $\mathrm{Vol}_{i,o,t} > 0$.
$\mathrm{Coverage}^{\mathcal{K}}_{i,o,t}$	$\sum_{k\in\mathcal{K}\setminus\{i,o\}} \mathrm{VehicleShare}_{i,o,k,t}$	Fraction (0–1)	`pair_candidate_vehicle_coverage`	Share of pair (i,o)'s realized indirect-route volume routed through a candidate in $\mathcal{K}\setminus\{i,o\}$; defined when total pair indirect volume is positive.
$\mathrm{VehicleHHI}_{i,o,t}$	$\sum_{k\in\mathcal{K}\setminus\{i,o\}} \left(\frac{\mathrm{VehicleShare}_{i,o,k,t}}{\mathrm{Coverage}^{\mathcal{K}}_{i,o,t}}\right)^2$	Fraction (0–1)	`pair_vehicle_hhi`	Herfindahl concentration among candidates in $\mathcal{K}\setminus\{i,o\}$ for ordered pair (i,o) on day t, after renormalizing their shares to sum to one; defined when candidate-routed indirect volume is positive.
Network and route-denominator controls				
$\mathrm{VolShare}_{k,t}$	$\frac{\mathrm{LegVol}^{\mathrm{in}}_{k,t} + \mathrm{LegVol}^{\mathrm{out}}_{k,t}}{\sum_\ell(\mathrm{LegVol}^{\mathrm{in}}_{\ell,t} + \mathrm{LegVol}^{\mathrm{out}}_{\ell,t})}$	Fraction (0–1)	`vol_share`	$\mathrm{LegVol}^{\mathrm{in}}_{k,t}$ and $\mathrm{LegVol}^{\mathrm{out}}_{k,t}$ are input-side and output-side route-leg USD volumes for k; ℓ indexes every token in the day-t network.
$\mathrm{RouteIncidence}^N_{k,t}$	$\frac{N^I_{k,t}}{N_t - N^{\mathrm{in}}_{k,t} - N^{\mathrm{out}}_{k,t}}$	Fraction (0–1)	`betweenness_centrality`	Fraction of day-t intent routes on which k is intermediate, excluding routes where k is the input or output endpoint. Superscripts identify each route role.
$\mathrm{RouteIncidence}^V_{k,t}$	$\frac{\mathrm{IVol}_{k,t}}{\mathrm{Vol}_t - \mathrm{Vol}^{\mathrm{in}}_{k,t} - \mathrm{Vol}^{\mathrm{out}}_{k,t}}$	Fraction (0–1)	`volume_weighted_betweenness`	USD-volume analogue of $\mathrm{RouteIncidence}^N_{k,t}$ using the same route universe and input/output-endpoint exclusions.
Vol_t	$\mathrm{DVol}_t + \mathrm{IVol}_t$	USD	`daily_all_route_volume_usd`	Total realized USD volume across direct and indirect route units on day t.
IVol_t		USD	`daily_indirect_route_volume_usd`	Total realized USD volume across indirect route units on day t.
$\mathrm{IndirectRouteShare}_t$	$\frac{\mathrm{IVol}_t}{\mathrm{Vol}_t}$	Fraction (0–1)	`indirect_route_share`	Fraction of day-t all-route USD volume executed through indirect routes.
LP capital measures				
$C_{k,t}$	$\sum_{p\in\mathcal{L}_{k,t}} \frac{\mathrm{Capital}_{p,t}}{m_p}$	USD	`vehicle_linked_capital_usd`	Candidate k's allocated share of validated accounting capital across every admitted protocol: full capital when k is the pool's only candidate token and one half when both pool tokens are candidates. This is deposited capital, not marginal depth.
$\mathrm{LPCapitalShare}_{k,t}$	$\frac{C_{k,t}}{\sum_{\ell\in\mathcal{K}} C_{\ell,t}}$	Fraction (0–1)	`lp_capital_share`	Candidate k's share of total candidate-linked deposited capital on day t.
$\mathrm{LogVehicleCapital}_{k,t}$	$\ln(1 + C_{k,t})$	Natural-log points	`log_vehicle_linked_capital`	Natural log of one plus $C_{k,t}$ measured in USD.
LP capital commonality measures				
$\mathrm{VehicleCapitalFactor}_{p,k,t}$	$\frac{\sum_{p'\in\mathcal{L}_{k,t}\setminus\{p\}} \Delta_1 \ln(\mathrm{Capital}_{p',t})}{\lvert\mathcal{L}_{k,t}\setminus\{p\}\rvert}$	Daily log-change points	`vehicle_capital_factor_loo`	Leave-one-out mean daily log deposited-capital change among other valid pools linked to candidate k; requires at least three other pools.
$\mathrm{MarketCapitalFactor}_{p,t}$	$\frac{\sum_{p'\in\mathcal{L}_t\setminus\{p\}} \Delta_1 \ln(\mathrm{Capital}_{p',t})}{\lvert\mathcal{L}_t\setminus\{p\}\rvert}$	Daily log-change points	`market_capital_factor_loo`	Leave-one-out mean daily log deposited-capital change across all other valid pools in $\mathcal{L}_t$; market-wide control for the vehicle factor.
Liquidity commonality measures				
$\mathrm{VehicleRouteShare}_{p,k,t}$	$\frac{\mathrm{SpokeIVol}_{p,k,t}}{\mathrm{PoolVol}_{p,t}}$	Fraction (0–1)	`pool_vehicle_route_share`	Fraction of pool p's day-t realized swap volume contributed by legs of indirect routes using k as the intermediate; defined for positive pool volume.
LP capital and return measures				
$L_{a,p,t}$		USD	`lp_active_capital_usd`	End-of-day active liquidity controlled by address a in pool p, reconstructed from position events and marked at day-t reference prices. Contract-managed positions are assigned to a controller only where the event trail permits look-through.
$w_{a,p,t}$	$\frac{L_{a,p,t}}{\sum_{a'} L_{a',p,t}}$	Fraction (0–1)	`lp_pool_capital_share`	Provider a's share of reconstructed active capital in pool p on day t.
$F^{\mathrm{LP}}_{a,p,t}$		USD per day	`lp_net_flow_usd`	Transaction-time USD value of day-t mint deposits less burn withdrawals for provider a in pool p, with fee collections excluded.
$\mathrm{LPFeeYield}_{a,p,t}$	$\frac{\mathrm{Fee}_{a,p,t}}{L_{a,p,t-1}}$	Daily fraction	`lp_fee_yield`	Day-t fees accrued to the position divided by lagged active capital; defined only for positive lagged capital.
$\mathrm{LVR}_{a,p,t}$	$\frac{V^{\mathrm{RB}}_{a,p,t} - V^{\mathrm{LP}}_{a,p,t}}{L_{a,p,t-1}}$	Daily fraction	`lp_lvr`	Underperformance of the opening LP inventory relative to its self-financing rebalanced benchmark, before fees, flows, and gas; defined for positive lagged capital.
$\mathrm{LPNetReturn}_{a,p,t}$	$\mathrm{LPFeeYield}_{a,p,t} - \mathrm{LVR}_{a,p,t} - \frac{G^{\mathrm{LP}}_{a,p,t}}{L_{a,p,t-1}}$	Daily fraction	`lp_net_return`	Fee yield less loss versus rebalancing and position-management gas cost, all scaled by lagged active capital.
$R^{\mathrm{other}}_{a,-p,t}$	$\frac{\sum_{p'\neq p} L_{a,p',t-1} R^{\mathrm{LP}}_{a,p',t}}{\sum_{p'\neq p} L_{a,p',t-1}}$	Daily return fraction	`lp_other_pool_return`	Lag-capital-weighted, fee-, flow-, and gas-excluded return on provider a's positions outside focal pool p; defined when outside lagged capital is positive.
$Z^{\mathrm{other}}_{a,-p,t}$	$\sum_{x\notin p} \omega_{a,x,-p,t-1} R_{x,t}$	Daily return fraction	`lp_predicted_other_pool_shock`	Shift-share return shock formed from provider a's lagged outside-pool token exposures and independent day-t token returns, excluding both focal-pool tokens.
$\mathrm{LPWealthShock}_{p,t}$	$\sum_a w_{a,p,t-1} Z^{\mathrm{other}}_{a,-p,t}$	Daily return fraction	`pool_lp_wealth_shock`	Lagged-capital-share-weighted predicted outside-pool shock of the providers supplying pool p before day-t flows.
$\mathrm{LPOverlap}_{p,p',t}$	$\sum_a \min\{w_{a,p,t}, w_{a,p',t}\}$	Fraction (0–1)	`lp_provider_overlap`	Common active-capital share attributable to the same resolved providers in pools p and p'; zero means no overlap and one means identical provider weights.
Routing-search efficiency measures				
$\mathrm{ChosenError}_r$	$10^4(O^{\mathrm{chosen}}_r - O^{\mathrm{real}}_r)/O^{\mathrm{real}}_r$	Basis points	`chosen_validation_error_bps`	Signed difference between the exact pre-transaction quote of realised route r and its observed token output. The construction sample requires an absolute error of at most one basis point; 0.1- and 0.01-basis-point subsets are prespecified.
$\lvert\mathrm{ChosenError}^{\max}_r\rvert$	$\max\{\lvert e^{(1)}_r\rvert, \lvert e^{(2)}_r\rvert, \lvert e^{(1\circ 2)}_r\rvert\}$	Basis points	`chosen_validation_max_abs_error_bps`	Maximum absolute error across the first realised leg, the second realised leg quoted at its observed input, and their composed path. The construction sample requires this measure to be at most one basis point; 0.1- and 0.01-basis-point subsets are prespecified.
$\mathrm{Regret}^{\mathrm{reach}}_r$	$10^4[\max\{O^{\mathrm{real}}_r, O^{k,\mathcal{V}_r}_r\} - O^{\mathrm{real}}_r]/O^{\mathrm{real}}_r$	Basis points	`within_reach_search_regret_bps`	Output gain from the best path through the realised vehicle k using only the venues observed on route r, relative to realised output at the same pre-transaction state.
$\mathrm{ReachIncrement}_r$	$10^4[\max\{O^{\mathrm{real}}_r, O^{k,\mathcal{V}}_r\} - \max\{O^{\mathrm{real}}_r, O^{k,\mathcal{V}_r}_r\}]/O^{\mathrm{real}}_r$	Basis points	`reach_increment_bps`	Additional output from expanding the same-vehicle search from route r's observed venues $\mathcal{V}_r$ to the admitted public venue set $\mathcal{V}$.
$\mathrm{PathChoiceIncrement}_r$	$10^4[\max\{O^{\mathrm{real}}_r, O^{\mathrm{public}}_r\} - \max\{O^{\mathrm{real}}_r, O^{k,\mathcal{V}}_r\}]/O^{\mathrm{real}}_r$	Basis points	`path_choice_increment_bps`	Additional output from allowing the admitted public frontier to change the intermediary or choose a direct path after public venue reach is already available.
$\mathrm{Regret}^{\mathrm{public}}_r$	$\mathrm{Regret}^{\mathrm{reach}}_r + \mathrm{ReachIncrement}_r + \mathrm{PathChoiceIncrement}_r$	Basis points	`public_path_regret_bps`	Total nonnegative output shortfall against the best admitted direct or two-leg public path at the same pre-transaction state.
$\mathrm{DirectOmission}_r$	$10^4[\max\{O^{\mathrm{real}}_r, O^{D,\mathcal{V}}_r\} - O^{\mathrm{real}}_r]/O^{\mathrm{real}}_r$	Basis points	`direct_omission_bps`	Output gain from the best admitted public direct path relative to the realised indirect route; undefined when no direct path is executable.
Route-cost opportunity measures				
$\mathrm{AnyIndirectAvailable}_{i,o,q,t}$	$\mathbf{1}_{\{\sum_{k\in\mathcal{K}\setminus\{i,o\}} I_{i,o,k,q,t} > 0\}}$	Indicator (0/1)	`any_indirect_available`	Equals one when at least one candidate in $\mathcal{K}\setminus\{i,o\}$ provides an executable indirect route for pair (i,o) at input notional q.
$\mathrm{DirectQuoteQuality}_{i,o,q,t}$	$\frac{O^D_{i,o,q,t}}{q}$	Output USD per input USD	`pair_direct_quote_quality`	Direct-route output ratio for cells with $D_{i,o,q,t} = 1$; undefined when the direct route is not executable.
$\mathrm{IndirectQuoteQuality}_{i,o,k,q,t}$	$\frac{O^I_{i,o,k,q,t}}{q}$	Output USD per input USD	`pair_indirect_quote_quality`	Indirect-route output ratio through candidate k for $(i,o) \in \mathcal{I}_{k,t,q}$; undefined when that route is not executable.
$C^{D,\mathrm{fee}}_{i,o,q,t}$	$1 - \frac{O^{D,\mathrm{fee}}_{i,o,q,t}}{q}$	Fraction	`pair_direct_fee_cost`	Direct-route loss attributable to historical pool fees when price impact is suppressed at the pre-quote marginal price.
$C^{D,\mathrm{impact}}_{i,o,q,t}$	$\frac{O^{D,\mathrm{fee}}_{i,o,q,t} - O^D_{i,o,q,t}}{q}$	Fraction	`pair_direct_price_impact_cost`	Additional direct-route output loss caused by moving through the realized liquidity curve, after holding the fee-only output fixed.
$C^{D,\mathrm{gas}}_{i,o,q,t}$	$\frac{G^D_{i,o,q,t}}{q}$	Fraction	`pair_direct_gas_cost`	Historical direct-route gas expenditure as a fraction of input USD notional.
$C^{I,\mathrm{fee}}_{i,o,k,q,t}$	$1 - \frac{O^{I,\mathrm{fee}}_{i,o,k,q,t}}{q}$	Fraction	`pair_indirect_fee_cost`	Two-leg indirect-route loss attributable to both pools' historical fees when price impact is suppressed at each pre-quote marginal price.
$C^{I,\mathrm{impact}}_{i,o,k,q,t}$	$\frac{O^{I,\mathrm{fee}}_{i,o,k,q,t} - O^I_{i,o,k,q,t}}{q}$	Fraction	`pair_indirect_price_impact_cost`	Additional two-leg indirect-route output loss caused by moving through both realized liquidity curves, after holding fee-only output fixed.
$C^{I,\mathrm{gas}}_{i,o,k,q,t}$	$\frac{G^I_{i,o,k,q,t}}{q}$	Fraction	`pair_indirect_gas_cost`	Historical two-leg indirect-route gas expenditure as a fraction of input USD notional.
$C^D_{i,o,q,t}$	$1 - \frac{O^D_{i,o,q,t}}{q} + \frac{G^D_{i,o,q,t}}{q}$	Fraction	`pair_all_in_direct_cost`	Direct quote-output loss, which includes pool fees and price impact, plus historical route gas as a fraction of input notional. Fee, price-impact, and gas contributions are retained separately and sum to this measure.
$C^I_{i,o,k,q,t}$	$1 - \frac{O^I_{i,o,k,q,t}}{q} + \frac{G^I_{i,o,k,q,t}}{q}$	Fraction	`pair_all_in_indirect_cost`	Indirect quote-output loss across both legs, which includes both pools' fees and price impact, plus historical route gas as a fraction of input notional. Fee, price-impact, and gas contributions are retained separately and sum to this measure.
$\Delta C^{D,\mathrm{all}}_{i,o,k,q,t}$	$C^I_{i,o,k,q,t} - C^D_{i,o,q,t}$	Fraction	`pair_all_in_direct_cost_advantage`	Indirect all-in cost minus direct all-in cost on common support; positive values favor direct execution and negative values favor indirect execution.
$\mathrm{DirectAvailable}_{k,t,q}$	$\frac{\lvert\mathcal{D}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$	Fraction (0–1)	`direct_available_share`	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $D_{i,o,q,t} = 1$. The un-suffixed data column fixes $q = \$10{,}000$.
$\mathrm{IndirectAvailable}_{k,t,q}$	$\frac{\lvert\mathcal{I}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$	Fraction (0–1)	`vehicle_available_share`	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $I_{i,o,k,q,t} = 1$, so both legs through k are executable.
$\mathrm{IndirectOnlyAvailable}_{k,t,q}$	$\frac{\lvert\mathcal{I}_{k,t,q}\setminus\mathcal{D}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$	Fraction (0–1)	`no_direct_vehicle_available_share`	Fraction of pairs with $D_{i,o,q,t} = 0$ and $I_{i,o,k,q,t} = 1$: no executable direct route but an executable route through k.
$\mathrm{DirectQuoteQuality}_{k,t,q}$	$\mathrm{median}_{\mathcal{D}_{k,t,q}} \left(O^D_{i,o,q,t}/q\right)$	Output USD per input USD	`direct_quote_quality_median`	Median direct-route output ratio $O^D_{i,o,q,t}/q$ over executable-direct pairs $(i,o) \in \mathcal{D}_{k,t,q}$.
$\mathrm{DirectCostAdvantage}_{k,t,q}$	$\mathrm{median}_{\mathcal{C}_{k,t,q}} \Delta C^D_{i,o,k,q,t}$	Fraction	`direct_cost_advantage_median`	Median $\Delta C^D_{i,o,k,q,t}$ over common-support pairs $(i,o) \in \mathcal{C}_{k,t,q}$; positive values favor the direct route and negative values favor the indirect route through k.
$\mathrm{AllInDirectCostAdvantage}_{k,t,q}$	$\mathrm{median}_{\mathcal{C}_{k,t,q}} \Delta C^{D,\mathrm{all}}_{i,o,k,q,t}$	Fraction	`all_in_direct_cost_advantage_median`	Median pair-level all-in direct cost advantage over common-support pairs; positive values favor direct execution after gas.
$\mathrm{IndirectBeatsDirect}_{k,t,q}$	$\frac{\lvert\mathcal{W}_{k,t,q}\rvert}{\lvert\mathcal{C}_{k,t,q}\rvert}$	Fraction (0–1)	`vehicle_beats_direct_share`	Fraction of $(i,o) \in \mathcal{C}_{k,t,q}$ for which $\Delta C^D_{i,o,k,q,t} < 0$, equivalently $O^I_{i,o,k,q,t} > O^D_{i,o,q,t}$.
$\mathrm{ThinDirectShare}_{k,t,q}$	$\frac{\lvert\mathcal{T}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$	Fraction (0–1)	`thin_direct_share`	Fraction of pairs with an executable direct quote but $O^D_{i,o,q,t}/q < 0.9$. This is a quote-quality proxy, not a direct measure of pool liquidity depth.
Stress and dynamic variables				
$\mathrm{CandidateStress}_{k,t}$	$\max\left\{-\frac{R_{k,t}}{\sigma^{(30)}_{k,t-1}}, 0\right\}$	Trailing-volatility standard deviations	`candidate_downside_stress`	Candidate-specific adverse return shock scaled by trailing volatility. The measure captures crashes for volatile candidates and downward depegs for stable candidates.
Stress_t	$\max\{-R^{\mathrm{WETH}}_t, 0\}$	Daily log-return fraction	`stress_downside`	Positive part of the negative day-t WETH log return.
$\mathrm{StressEvent}_t$	$\mathbf{1}_{\{\mathrm{Stress}_t \ge 0.08\}}$	Indicator (0/1)	`stress_event_8pct`	Equals one when Stress_t is at least 0.08 on day t.
$\Delta_\tau \mathrm{VehicleShare}_{k,t}$	$\mathrm{VehicleShare}_{k,t} - \mathrm{VehicleShare}_{k,t-\tau}$	Fraction points	`delta_bridge_share_t7`	τ-day change for vehicle k ending on day t. The displayed data column is the $\tau = 7$ instance; the observations table also constructs $\tau \in \{1, 7, 30, 120\}$ by exact calendar date.
Reference-price support				
$\mathrm{CEXRef}_{x,t}$	$\mathbf{1}_{\{x\in\mathcal{K}^{\mathrm{CEX}},\ t\in[t^{\mathrm{first}}_x, t^{\mathrm{last}}_x]\}}$	Indicator (0/1)	`cex_reference_supported`	Equals one only for an exact token address in the published 43-pair Uniswap–Binance comparison and between that pair's first and last observations in the package's 1-in-10,000 minute sample. Absence is unsupported and never means that the token was unlisted.
Persistence and displacement measures				
$\mathrm{Incumbent}_{i,o,k,t}$	$\mathbf{1}_{\{k = k^\star_{i,o,t}\}}$	Indicator (0/1)	`incumbent_vehicle`	Equals one when candidate k is the trailing-30-day incumbent for ordered pair (i,o); the ranking window ends at $t-1$.
$\mathrm{ChallengerCostEdge}_{i,o,q,t}$	$C^I_{i,o,k^\star,q,t} - C^I_{i,o,h^\star,q,t}$	Fraction	`challenger_cost_edge`	All-in cost advantage of the best executable challenger over the incumbent vehicle at input notional q; positive values favor the challenger.
$\mathrm{VehicleSwitch}_{i,o,q,t,\tau}$	$\mathbf{1}\left\{\begin{array}{c}\mathrm{VehicleShare}_{i,o,h^\star,t+\tau} \\ > \mathrm{VehicleShare}_{i,o,k^\star,t+\tau}\end{array}\right\}$	Indicator (0/1)	`vehicle_switch`	Equals one when the day-t best challenger has a larger pair-level vehicle share than the day-t incumbent at exact calendar horizon $t+\tau$.
Execution-architecture measures				
$\mathrm{DirectConstraint}^{\mathrm{pre}}_{i,o,q}$	$1 - \frac{\sum_{u\in\mathcal{T}^{\mathrm{V3}}_{\mathrm{pre}}} D_{i,o,q,u}}{\lvert\mathcal{T}^{\mathrm{V3}}_{\mathrm{pre}}\rvert}$	Fraction (0–1)	`pre_v3_direct_constraint`	Fraction of the fixed 180-day pre-V3 window in which pair (i,o) lacks an executable direct route at notional q; higher values mean a more constrained pre-architecture direct market.
$\sigma^{\mathrm{pre}}_{i,o}$	$\mathrm{sd}_{u\in\mathcal{T}^{\mathrm{V3}}_{\mathrm{pre}}}(R_{i,o,u})$	Daily log-return standard deviation	`pre_v3_pair_volatility`	Sample standard deviation of the ordered endpoint-pair log return over the fixed 180-day pre-V3 window; computed only from information dated before V3 activation.
$\eta^{\mathrm{Band}}_{p,t,b,d}$	$\frac{\mathrm{BandDepth}_{p,t,b,d}}{C_{p,t}}$	USD directional band depth per USD validated deposited capital	`pool_band_depth_capital_efficiency`	Fee-inclusive executable dollar depth in direction d inside the symmetric b price band divided by contemporaneous validated deposited capital. The primary band is $b = 0.02$; $b \in \{0.01, 0.10\}$ is retained for robustness. The row retains direction, band, invariant family and state generation.
$\mathrm{PostV3}_t$	$\mathbf{1}_{\{t \ge t^{\mathrm{V3}}_0\}}$	Indicator (0/1)	`post_v3`	Equals one on and after the Uniswap V3 activation date.
V4 settlement implementation measures				
$\mathrm{Transfer}_{r,k}$	$\mathbf{1}_{\{r\in\mathcal{R}^{\mathrm{transfer}}_{k,w}\}}$	Indicator (0/1)	`has_matching_transfer`	Equals one when route unit r's transaction receipt contains an ERC-20 Transfer log matching intermediate vehicle k.
$\mathrm{V4}_r$	$\mathbf{1}_{\{r\in\mathcal{R}^4_g\}}$	Indicator (0/1)	`v4_route`	Equals one when matched route unit r executes on Uniswap V4.
$\mathrm{V4RouteShare}_g$	$\frac{\lvert\mathcal{R}^4_g\rvert}{\lvert\mathcal{R}^3_g\rvert + \lvert\mathcal{R}^4_g\rvert}$	Fraction (0–1)	`v4_route_share`	Fraction of receipt-audited route units in comparison cell g executed on V4; reported alongside transfer incidence to distinguish route use from token movement.
$\mathrm{PreV4IndirectShare}_{i,o}$	$\frac{\sum_{u\in\mathcal{T}^{\mathrm{V4}}_{i,o,\mathrm{pre}}} \mathrm{IndirectRouteShare}_{i,o,u}}{\lvert\mathcal{T}^{\mathrm{V4}}_{i,o,\mathrm{pre}}\rvert}$	Fraction (0–1)	`pre_v4_pair_indirect_route_share`	Mean pair-level indirect-route share over pair (i,o)'s positive-volume days in the fixed 180-day pre-V4 window, requiring at least 30 such days.
$\mathrm{PostV4}_t$	$\mathbf{1}_{\{t \ge t^{\mathrm{V4}}_0\}}$	Indicator (0/1)	`post_v4`	Equals one on and after the Ethereum V4 activation date.
$M_{r,k}$		USD	`physical_vehicle_movement_usd`	Sum of absolute, route-attributed ERC-20 transfers of intermediate token k between distinct nonzero addresses in route unit r, with duplicate logs removed and each amount valued at the transaction-time reference price.
$\mathrm{SettlementIntensity}_{r,k}$	$\frac{M_{r,k}}{\mathrm{GrossLegVol}_{r,k}}$	USD transferred per gross vehicle-leg USD	`physical_settlement_intensity`	Physical intermediate-token movement per unit of gross intermediate-token notional across the two economic route legs; defined for positive gross vehicle-leg volume.
$\mathrm{VehicleTurnover}_{k,t}$	$\frac{\mathrm{IVol}_{k,t}}{C_{k,t}}$	USD vehicle volume per USD deposited capital per day	`vehicle_capital_turnover`	Realized indirect-route volume through candidate k divided by its allocated vehicle-linked deposited capital, defined for positive $C_{k,t}$.
$\mathrm{VehicleRouteExposure}^{\mathrm{pre}}_{p,k}$	$\frac{\sum_{u\in\mathcal{T}^{\mathrm{V4}}_{\mathrm{pre}}} \mathrm{SpokeIVol}_{p,k,u}}{\sum_{u\in\mathcal{T}^{\mathrm{V4}}_{\mathrm{pre}}} \mathrm{PoolVol}_{p,u}}$	Fraction (0–1)	`pre_v4_pool_vehicle_route_exposure`	Fraction of pool p's realized swap volume in the fixed 180-day pre-V4 window attributable to legs of indirect routes using k as the intermediate; defined for positive pre-period pool volume.
$\mathrm{TransferIncidence}_{k,w}$	$\frac{\lvert\mathcal{R}^{\mathrm{transfer}}_{k,w}\rvert}{\lvert\mathcal{R}_{k,w}\rvert}$	Fraction (0–1)	`settlement_transfer_incidence`	Fraction of receipt-audited route units $r \in \mathcal{R}_{k,w}$ containing an ERC-20 Transfer log for intermediate vehicle k.
$\mathrm{ReceiptCount}_{k,w}$	$\lvert\mathcal{R}_{k,w}\rvert$	Route-unit count	`settlement_receipt_count`	Number of receipt-audited matched route units for vehicle k in UTC week w.